Vijay Mohanram Iyer

Tennesseeallee 20, Karlsruhe, Baden-Württemberg, Germany

iyervijay99@gmail.com +49-17667345305 Vijay's_Portfolio Vijay Iyer

SUMMARY

Graduate engineer specializing in intelligent safety systems and health applications powered by computer vision and machine learning. Skilled in transforming advanced algorithms including deep learning architectures and large language models into practical, user-centered tools that enhance security and promote human well-being.

PROFESSIONAL EXPERIENCE

1. FZI *August 2025 – November 2025*
   Research Assistant

   - To enhance DeepFake detection framework with dynamic ROI tracking using 3D facemesh and computer vision for landmark localization.
   - To develop interactive UI visualizing multiple ROIs with artifact energy mapping via advanced data visualization and UI/UX design principles..
   - Fine-tune models using transfer learning, hyperparameter optimization, and ensemble methods for robust detection and research paper publication.

2. TecoLab *March 2023 – September 2025*
   Working Student

   - ML4Print: Classification based on printer-specific characteristics and paper substrate type using feature extraction and supervised ML classifiers.
   - Heat simulation: utilized regression models and CFD data integration to simulate the thermal behavior of liquids within industrial valves, thereby enhancing predictive maintenance capabilities and system reliability.
   - Sensor optimization for open-earables with TinyML and quantized neural networks for on-device inference.

3. Access@KIT *March 2023 – May 2023*
   Working Student

   - Designed and implemented a bi-directional LSTM network for text-to-speech synthesis, optimizing acoustic feature prediction to capture temporal dependencies and prosody for natural, intelligible speech.
   - Performed data preprocessing including phoneme alignment and feature extraction, and evaluated model performance with iterative tuning to enhance digital accessibility for visually impaired users.

4. Accenture India Pvt. Ltd. *February 2022 – April 2022*
   Associate Software Engineer

   - Maintained IBM Mainframe systems with batch job scheduling and performance monitoring scripts.

5. Accur Digitus *January 2020 – May 2020*
Web Developer Intern

- Developed React.js web applications with component-based architecture and state management.
- Implemented RESTful APIs using JSON and asynchronous fetch for front-end/back-end integration.

EDUCATION

M.Sc. in Electrical Engineering and Information Technology *May 2022 – July 2025*
Karlsruhe Institute of Technology
GPA: 2.3

Bachelor of Engineering in Electronics Engineering *August 2017 – May 2021*
University of Mumbai, India
GPA: 2.8

PROJECTS

1. **Attention Mechanism and Multi-ROI Artifact Recognition for DeepFake Detection Using rPPG (FZI)** *February 2025 – July 2025*
   This Master Thesis aimed to develop an end-to-end deep learning pipeline using rPPG signals for artifact evaluation. Proposed a Fusion Model combining Vision Transformer and CNN features via a custom Fusion-Head for DeepFake classification. Benchmarked under real-world conditions.
2. **CamCussion: Eye Tracking Software (Zeiss Innovation Hub)** *December 2023 – July 2024*
   Utilized computer-vision to analyze pupil dilation and saccadic eye movements in real time to track and assess eye behavior, contributing to accurate concussion diagnosis.
3. **Self-Driving Car using LIDAR** *September 2021 – February 2022*
   Developed a solar-powered autonomous vehicle prototype using 360° LIDAR for reliable obstacle detection and safe navigation. Implemented on Arduino with custom chassis and differential drive, aimed at enhancing safety in urban mobility applications.
4. **Real-Time Car Accident Alert System** *January 2021 – june 2021*
   Developed an embedded vehicle accident warning system to automatically detect crashes, send precise location data, and alert emergency services, family, and friends to improve response times.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, SQL, React.js, Node.js, HTML, CSS
- **Libraries/Frameworks:** OpenCV, Qt, TensorFlow, PyTorch, pandas, NumPy, Matplotlib, scikit-learn, ROS, PCL
- **ADAS:** Sensor fusion, camera and LiDAR calibration, perception algorithms, path planning, object detection & tracking, ADAS prototyping
- **Tools:** Git, Docker, MATLAB, Linux, Carla, conda, LaTeX
- **Spoken Languages:** English C1, German A2 (Learning)

SOFT SKILLS

- **Technical Leadership:** Project Development, Cross-functional Collaboration, Innovative mindset.
- **Problem Solving:** Critical Thinking, Performance Optimization, Debugging, Balancing and Prioritizing tasks to meet goals.
- **Communication:** Active Listening, Documenting, Knowledge Transfer.